

Perturbations of Spherically Symmetric Spacetimes in $f(R) = R + \alpha R^2$ Gravity: Personal Notes

Anant (compiled on Overleaf)

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1 Introduction

In $f(R) = R + \alpha R^2$ gravity, the parameter α determines the strength of the R^2 term, which modifies the standard General Relativity theory. Positive α values lead to modifications in stellar models, while negative α values can introduce issues with gravitational mass and stability.

1.1 Elaboration:

- **Positive α :** When alpha is positive, the scalar curvature does not drop to zero at the star surface, and there is a gravitational sphere around the star that contributes to its observed mass. This leads to an increase in the star's radius, especially for stars near their maximum mass.
- **Negative α :** Negative α values can cause the scalar curvature to oscillate, and the gravitational sphere's contribution to mass can increase indefinitely, potentially questioning the existence of such a sphere.
- **Stability:** Studies suggest that the parameter α in $f(R) = R + \alpha R^2$ gravity needs to be small (e.g., $\alpha < 2.4 \times 10^8 \text{ cm}^2$) to ensure stellar stability.
- **Inflation:** The R^2 term in $f(R) = R + \alpha R^2$ can influence the height of the effective inflation potential during inflation, allowing for a suppression of the tensor-to-scalar ratio without affecting the spectrum of scalar perturbations.

1.2 Comparison to General Relativity:

In General Relativity (where $\alpha = 0$), the scalar curvature goes to zero at the surface of a star, and the gravitational mass is solely determined by the mass within the star's surface. In contrast, $f(R) = R + \alpha R^2$ introduces additional terms related to the spacetime curvature, influencing the gravitational mass and radius of stars.

2 Field Equations in Jordan Frame

Varying the action with respect to the metric $g_{\mu\nu}$, the field equations for general $f(R)$ gravity in vacuum are:

$$f'(R)R_{\mu\nu} - \frac{1}{2}f(R)g_{\mu\nu} + (g_{\mu\nu}\square - \nabla_\mu\nabla_\nu)f'(R) = 0 \quad (1)$$

where:

- $f'(R) = \frac{df(R)}{dR}$,
- $\square = g^{\mu\nu}\nabla_\mu\nabla_\nu$

Contracting with the inverse metric $g^{\mu\nu}$, we get the trace equation:

$$f'(R)R - 2f(R) + 3\square f'(R) = 0 \quad (2)$$

With our model $f(R) = R + \alpha R^2$, $f'(R) = 1 + 2\alpha R$, so the field equations and the trace equation look like:

$$(1 + 2\alpha R)R_{\mu\nu} - \frac{1}{2}(R + \alpha R^2)g_{\mu\nu} + 2\alpha(g_{\mu\nu}\square - \nabla_\mu \nabla_\nu)R = 0$$

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + 2\alpha R R_{\mu\nu} - \frac{1}{2}\alpha R^2 g_{\mu\nu} + 2\alpha(g_{\mu\nu}\square - \nabla_\mu \nabla_\nu)R = 0 \quad (3)$$

and

$$\square R - \frac{1}{6\alpha}R = 0 \quad (4)$$

This is a Klein-Gordon equation for the Ricci scalar R with a mass $m^2 = 1/6\alpha$.

3 Linearization of $R + \alpha R^2$ Gravity Around Schwarzschild Background in the Jordan Frame

3.1 Introduction

In this section, we study the linearized perturbations of $f(R) = R + \alpha R^2$ gravity about a Schwarzschild background, working in the Jordan frame. The aim is to derive the linearized field equations satisfied by small perturbations around Schwarzschild spacetime.

Unlike General Relativity, where the linearized equations around Schwarzschild involve only metric perturbations, in $R + \alpha R^2$ gravity, the theory has an additional propagating scalar degree of freedom associated with the Ricci scalar R . Understanding the behavior of these perturbations is essential for computing quasi-normal modes (QNMs) in $R + \alpha R^2$ gravity.

3.2 Linearization Around Schwarzschild

The action for $R + \alpha R^2$ gravity, modulo the boundary terms, in the Jordan frame is given by

$$S_{EH} = \int d^4x \sqrt{-g}(R + \alpha R^2) \quad (5)$$

Varying the action with respect to $g_{\mu\nu}$ yields the field equations (3). We consider perturbations around a Schwarzschild background, whose metric is:

$$ds^2 = -f(r)dt^2 + \frac{dr^2}{f(r)} + r^2(d\theta^2 + \sin^2\theta d\phi^2), \quad (6)$$

with

$$f(r) = 1 - \frac{2M}{r}.$$

The background Ricci tensor and Ricci scalar satisfy:

$$R_{\mu\nu}^{(0)} = 0, \quad R^{(0)} = 0, \quad (7)$$

where the superscript (0) denotes background quantities.

We introduce small perturbations to the metric:

$$g_{\mu\nu} = g_{\mu\nu}^{(0)} + h_{\mu\nu}, \quad (8)$$

and correspondingly, the Ricci scalar perturbation is:

$$R = \delta R, \quad (9)$$

since $R^{(0)} = 0$.

Expanding the field equations to linear order, we analyze each term separately:

- The terms $R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}$ linearize to $\delta R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}^{(0)}\delta R$.
- The term $2\alpha RR_{\mu\nu}$ becomes zero at linear order because both $R^{(0)} = 0$ and $R_{\mu\nu}^{(0)} = 0$.
- The term $-\frac{1}{2}\alpha R^2 g_{\mu\nu}$ is quadratic in δR , and hence vanishes at linear order.
- The term $-2\alpha\nabla_\mu\nabla_\nu R$ linearizes to:

$$-2\alpha\nabla_\mu\nabla_\nu\delta R. \quad (10)$$

- The term $+2\alpha g_{\mu\nu}\square R$ linearizes to:

$$2\alpha g_{\mu\nu}^{(0)}\square\delta R, \quad (11)$$

where $\square$ the d'Alembertian.

Therefore, the linearized field equations become:

$$\delta R_{\mu\nu} - \frac{1}{2}\delta R g_{\mu\nu}^{(0)} + 2\alpha g_{\mu\nu}^{(0)}\square\delta R - 2\alpha\nabla_\mu\nabla_\nu\delta R = 0. \quad (12)$$

Taking the trace of this equation with $g^{(0)\mu\nu}$, we find:

$$g^{(0)\mu\nu}\delta R_{\mu\nu} - 2\delta R + 6\alpha\square\delta R = 0,$$

or, equivalently,

$$\square\delta R - \frac{1}{6\alpha}\delta R = 0 \quad (13)$$

Thus, δR satisfies the massive Klein-Gordon equation on the Schwarzschild background.

4 Scalar Perturbations: Solving the Massive Klein-Gordon Equation in Schwarzschild Background

We now solve the massive Klein-Gordon equation for the Ricci scalar perturbation δR in the Schwarzschild background:

$$\square\delta R - m^2\delta R = 0, \quad (14)$$

where $\square$ is the d'Alembertian operator and $m^2 = \frac{1}{6\alpha}$ is the mass of the scalar field δR .

4.1 Explicit Form of the Wave Operator

The Schwarzschild metric is given by:

$$ds^2 = -f(r) dt^2 + \frac{dr^2}{f(r)} + r^2(d\theta^2 + \sin^2 \theta d\phi^2), \quad (15)$$

with

$$f(r) = 1 - \frac{2M}{r}. \quad (16)$$

The general expression for the d'Alembertian acting on a scalar field ϕ is:

$$\square \phi = \frac{1}{\sqrt{-g}} \partial_\mu (\sqrt{-g} g^{\mu\nu} \partial_\nu \phi), \quad (17)$$

where g is the determinant of the metric tensor.

For the Schwarzschild metric, the determinant is:

$$g = \det(g_{\mu\nu}) = -r^4 \sin^2 \theta, \quad (18)$$

thus

$$\sqrt{-g} = r^2 \sin \theta. \quad (19)$$

Substituting into the general formula, the wave operator becomes:

$$\square \delta R = \frac{1}{r^2 \sin \theta} \partial_\mu (r^2 \sin \theta g^{\mu\nu} \partial_\nu \delta R). \quad (20)$$

Expanding term-by-term, we find:

$$\square \delta R = -\frac{1}{f(r)} \partial_t^2 \delta R + \frac{1}{r^2} \partial_r (r^2 f(r) \partial_r \delta R) + \frac{1}{r^2} \Delta_{S^2} \delta R, \quad (21)$$

where Δ_{S^2} is the Laplacian on the unit 2-sphere:

$$\Delta_{S^2} \equiv \frac{1}{\sin \theta} \partial_\theta (\sin \theta \partial_\theta) + \frac{1}{\sin^2 \theta} \partial_\phi^2. \quad (22)$$

4.2 Separation of Variables

We propose a separation of variables ansatz for the perturbation:

$$\delta R(t, r, \theta, \phi) = e^{-i\omega t} Y_{\ell m}(\theta, \phi) \frac{\Psi(r)}{r}, \quad (23)$$

where:

- ω is the complex oscillation frequency,
- $Y_{\ell m}(\theta, \phi)$ are the spherical harmonics satisfying

$$\Delta_{S^2} Y_{\ell m} = -\ell(\ell + 1) Y_{\ell m}, \quad (24)$$

- $\Psi(r)$ is the radial function to be determined.

Substituting this ansatz into the wave equation will allow us to reduce the problem to an ordinary differential equation (ODE) for $\Psi(r)$.

5 Linearization of the Trace Equation around Schwarzschild Background in the Jordan Frame

In this section, we study the linear perturbations of the trace equation in the Jordan frame, specifically around the Schwarzschild solution. We consider the model $f(R) = R + \alpha R^2$, for which the trace equation reduces to a massive scalar field equation:

$$\square \delta R - m^2 \delta R = 0 \quad (25)$$

where $\square$ is the D'Alembertian operator constructed using the Schwarzschild background metric. We seek solutions that are separable in time, radial, and angular parts, and we proceed to derive the resulting radial equation governing the perturbations.

5.1 Derivation of the Radial Equation

We begin by assuming a separable ansatz for the perturbation:

$$\delta R(t, r, \theta, \phi) = e^{-i\omega t} Y_{\ell m}(\theta, \phi) \frac{\Psi(r)}{r} \quad (26)$$

where $Y_{\ell m}(\theta, \phi)$ are the standard spherical harmonics.

The Schwarzschild metric is given by:

$$ds^2 = -f(r) dt^2 + \frac{dr^2}{f(r)} + r^2 (d\theta^2 + \sin^2 \theta d\phi^2) \quad (27)$$

with

$$f(r) = 1 - \frac{2M}{r} \quad (28)$$

Expanding the D'Alembertian operator on a spherically symmetric background, we have:

$$\square \delta R = -\frac{1}{f(r)} \partial_t^2 \delta R + \frac{1}{r^2} \partial_r (r^2 f(r) \partial_r \delta R) + \frac{1}{r^2} \Delta_{S^2} \delta R \quad (29)$$

where Δ_{S^2} is the Laplacian on the two-sphere.

Let us compute each term separately:

Time Derivative Term: Using the ansatz,

$$\partial_t^2 \delta R = -\omega^2 e^{-i\omega t} Y_{\ell m} \frac{\Psi(r)}{r} \quad (30)$$

thus,

$$-\frac{1}{f(r)} \partial_t^2 \delta R = \frac{\omega^2}{f(r)} e^{-i\omega t} Y_{\ell m} \frac{\Psi(r)}{r} \quad (31)$$

Angular Laplacian Term: Since spherical harmonics satisfy

$$\Delta_{S^2} Y_{\ell m} = -\ell(\ell + 1) Y_{\ell m} \quad (32)$$

we get

$$\frac{1}{r^2} \Delta_{S^2} \delta R = -\frac{\ell(\ell + 1)}{r^3} e^{-i\omega t} Y_{\ell m} \Psi(r) \quad (33)$$

Radial Derivative Term: First, compute the radial derivative:

$$\partial_r \delta R = e^{-i\omega t} Y_{\ell m} \left(\frac{\Psi'(r)}{r} - \frac{\Psi(r)}{r^2} \right) \quad (34)$$

thus,

$$r^2 f(r) \partial_r \delta R = e^{-i\omega t} Y_{\ell m} (r f'(r) \Psi'(r) - f(r) \Psi(r)) \quad (35)$$

and further,

$$\partial_r (r^2 f(r) \partial_r \delta R) = e^{-i\omega t} Y_{\ell m} (r f'(r) \Psi'(r) + r f(r) \Psi''(r) - f'(r) \Psi(r)) \quad (36)$$

thus,

$$\frac{1}{r^2} \partial_r (r^2 f(r) \partial_r \delta R) = e^{-i\omega t} Y_{\ell m} \frac{1}{r^2} (r f'(r) \Psi'(r) + r f(r) \Psi''(r) - f'(r) \Psi(r)) \quad (37)$$

Combining All Terms: Substituting all pieces into the trace equation, we obtain:

$$\square \delta R = e^{-i\omega t} Y_{\ell m} \left[\frac{\omega^2}{f(r)} \frac{\Psi(r)}{r} + \frac{1}{r^2} (r f'(r) \Psi'(r) + r f(r) \Psi''(r) - f'(r) \Psi(r)) - \frac{\ell(\ell+1)}{r^3} \Psi(r) \right] \quad (38)$$

Since the prefactor $e^{-i\omega t} Y_{\ell m}$ is nonzero, the radial equation simplifies to:

$$\frac{\omega^2}{f(r)} \frac{\Psi(r)}{r} + \frac{1}{r^2} [r f'(r) \Psi'(r) + r f(r) \Psi''(r) - f'(r) \Psi(r)] - \frac{\ell(\ell+1)}{r^3} \Psi(r) - m^2 \Psi(r) = 0 \quad (39)$$

Summary: In this subsection, we derived the radial perturbation equation for the trace fluctuation δR around a Schwarzschild background in the Jordan frame of $f(R) = R + \alpha R^2$ gravity. The next step will be to recast this radial equation into a Schrödinger-like form to study the quasi-normal modes.

6 Setting up the Modified Regge-Wheeler Equations

Here we introduce the tortoise coordinate and simplify (39), so that it looks like a Schrodinger equation with a potential.

6.1 Tortoise coordinate

The tortoise coordinate r_* is defined by

$$dr_* = \frac{dr}{f(r)} \quad (40)$$

so that

$$\frac{d}{dr} = \frac{1}{f(r)} \frac{d}{dr_*}. \quad (41)$$

Thus,

$$\frac{d^2}{dr^2} = \frac{d}{dr} \left(\frac{d}{dr} \right) = \frac{d}{dr} \left(\frac{1}{f(r)} \frac{d}{dr_*} \right) = -\frac{f'(r)}{f^2(r)} \frac{d}{dr_*} + \frac{1}{f^2(r)} \frac{d^2}{dr_*^2} \quad (42)$$

6.2 Radial Equation in Tortoise Coordinate

Rewrite the radial equation (39), multiplying by r to eliminate denominators as much as possible:

$$\frac{\omega^2}{f(r)}\Psi + \frac{1}{r} [rf'(r)\Psi' + rf(r)\Psi'' - f'(r)\Psi] - \frac{l(l+1)}{r^2}\Psi - m^2\Psi = 0$$

Simplify:

$$\omega^2 \frac{\Psi}{f(r)} + f'(r)\Psi' + f(r)\Psi'' - \frac{f'(r)}{r}\Psi - \frac{l(l+1)}{r^2}\Psi - m^2\Psi = 0$$

Now rearranging:

$$f(r)\Psi'' + f'(r)\Psi' + \left(\omega^2 \frac{1}{f(r)} - \frac{f'(r)}{r} - \frac{l(l+1)}{r^2} - m^2 \right) \Psi = 0 \quad (43)$$

Now translate to tortoise coordinate:

Using (41) and (42), (43) can be written as:

$$\frac{d^2\Psi}{dr_*^2} + [\omega^2 - V_{\text{eff.}}] \Psi = 0, \quad (44)$$

where

$$V_{\text{eff.}} = f(r) \left(\frac{f'(r)}{r} + \frac{l(l+1)}{r^2} + m^2 \right) \quad (45)$$

is the effective potential.